A UNIFORM BOUND ON THE DEGREES IN WHICH THE HIGHER DIRECT IMAGES VANISH, STABLE UNDER ARBITRARY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- An affine etale cover of a quasi-compact algebraic space, and base change of pushforwards along it. Properties of Spaces, labels spaces-properties-lemma-quasi-compact-affine-cover (invoked at proof line 1485) and spaces-properties-lemma-pushforward-etale-base-change-modules (invoked at proof line 1493). They reduce the statement to the case where the target Y is an affine scheme, since the higher direct images restrict to those of the base changed morphism.
- Quasi-coherence of the higher direct images along a quasi-compact quasi-separated morphism. Same chapter spaces-cohomology.tex, label lemma-higher-direct-image (invoked at proof line 1500). With an affine target Y' it makes $R^if'_*\mathcal{F}'$ a quasi-coherent module whose global sections compute $H^i(X', \mathcal{F}')$.
- The Leray identification and vanishing on an affine space. Cohomology on Sites, label sites-cohomology-lemma-apply-Leray (invoked at proof line 1503) and Same chapter spaces-cohomology.tex, label proposition-vanishing (invoked at proof line 1506). Together they turn the vanishing of the higher direct images into the vanishing of $H^i(X', \mathcal{F}')$ over an affine base.
- The collection of covers and degrees attached to a quasi-separated space. Same chapter spaces-cohomology.tex, label lemma-vanishing-quasi-separated (invoked at proof lines 1510 and 1515). Its choices for X, base changed to X', give the vanishing $H^i(X', \mathcal{F}') = 0$ for $i \geq \max(p + d_p)$, which is the bound asserted.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0720: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 339–339. Navigation pointer for \ref{lemma-higher-direct-image} used inside the counted span (line 1500); the argument that consumes it is retained in the span.

Source reference. Original lemma 072C: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1385–1385. Navigation pointer for \ref{lemma-vanishing-quasi-separated} used inside the counted span (line 1510); the argument that consumes it is retained in the span.

Source reference. Original proposition 072B: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1333–1333. Navigation pointer for \ref{proposition-vanishing} used inside the counted span (line 1506); the argument that consumes it is retained in the span.

Source reference. Original lemma 0733: sources/stacks-chow/tags.txt, source lines 9047–9047. Navigation pointer for \ref{sites-cohomology-lemma-apply-Leray} used inside the counted span (line 1503); the argument that consumes it is retained in the span.

Source reference. Original lemma 03LX: sources/stacks-chow/tags.txt, source lines 4729–4729. Navigation pointer for \ref{spaces-properties-lemma-pushforward-etale-base-change-modules} used inside the counted span (line 1493); the argument that consumes it is retained in the span.

Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Navigation pointer for \ref{spaces-properties-lemma-quasi-compact-affine-cover} used inside the counted span (line 1485); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume that*

(1) *f is quasi-compact and quasi-separated, and*
(2) *Y is quasi-compact.*

Then there exists an integer $n(X \to Y)$ such that for any algebraic space Y', any morphism $Y' \to Y$ and any quasi-coherent sheaf $\mathcal{F}'$ on $X' = Y' \times_Y X$ the higher direct images $R^i f'_* \mathcal{F}'$ are zero for $i \geq n(X \to Y)$.

Proof. Let $V \to Y$ be a surjective étale morphism where V is an affine scheme, see Properties of Spaces, Lemma 03H6. Suppose we prove the result for the base change $f_V : V \times_Y X \to V$. Then the result holds for f with $n(X \to Y) = n(X_V \to V)$. Namely, if $Y' \to Y$ and $\mathcal{F}'$ are as in the lemma, then $R^i f'_* \mathcal{F}'|_{V \times_Y Y'}$ is equal to $R^i f'_{V,*} \mathcal{F}'|_{X'_V}$ where $f'_V : X'_V = V \times_Y Y' \times_Y X \to V \times_Y Y' = Y'_V$, see Properties of Spaces, Lemma 03LX. Thus we may assume that Y is an affine scheme.

Moreover, to prove the vanishing for all $Y' \to Y$ and $\mathcal{F}'$ it suffices to do so when Y' is an affine scheme. In this case, $R^i f'_* \mathcal{F}'$ is quasi-coherent by Lemma 0720. Hence it suffices to prove that $H^i(X', \mathcal{F}') = 0$, because $H^i(X', \mathcal{F}') = H^0(Y', R^i f'_* \mathcal{F}')$ by Cohomology on Sites, Lemma 0733 and the vanishing of higher cohomology of quasi-coherent sheaves on affine algebraic spaces (Proposition 072B).

Choose $U \to X$, d, $V_p \to U_p$ and d_p as in Lemma 072C. For any affine scheme Y' and morphism $Y' \to Y$ denote $X' = Y' \times_Y X$, $U' = Y' \times_Y U$, $V'_p = Y' \times_Y V_p$. Then $U' \to X'$, $d' = d$, $V'_p \to U'_p$ and $d'_p = d$ is a collection of choices as in Lemma 072C for the algebraic space X' (details omitted). Hence we see that $H^i(X', \mathcal{F}') = 0$ for $i \geq \max(p + d_p)$ and we win. □